

LLD-DP-001 DATA PIPELINE & STORAGE

Low Level Design — Data Pipeline & Storage

LLD	DP-001	v1.0	AWS	CONFIDENTIAL
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Kinesis | Timestream | S3 | DynamoDB | OpenSearch | Aurora
SpaceNet-IT | HLD Ref: SPACENET-HLD-001 | June 2026

1. Scope & Purpose

This LLD defines the detailed design of the Data Pipeline and Storage tier. It covers real-time streaming with Kinesis, time-series storage in Timestream, raw frame archival in S3, fleet state management in DynamoDB, log analytics in OpenSearch, and mission planning in Aurora PostgreSQL. It includes data models, retention policies, and backup strategies.

2. Kinesis Data Streams

2.1 Stream Configuration

Stream Name	Shards	Retention	Throughput	Destination
spacenet-telemetry-hk	10	24 hours	10 MB/s in, 20 MB/s out	Timestream Lambda, OpenSearch Firehose, Twitter
spacenet-telemetry-science	5	168 hours (7 days)	5 MB/s in, 10 MB/s out	S3 Firehose, Science processing Lambda
spacenet-telemetry-events	4	24 hours	2 MB/s in, 4 MB/s out	EventBridge Pipes, SNS Lambda, DynamoDB L
spacenet-raw-frames	8	24 hours	20 MB/s in, 20 MB/s out	S3 Firehose (WORM archive)

2.2 Partition Key Strategy

Stream	Partition Key	Strategy
HK Telemetry	HK-{spacecraft_id}-{vcid}	Spread by satellite + VC — avoid hot shard
Science Data	SCI-{spacecraft_id}	One shard per satellite science stream
Events / Alerts	EVT-{satellite_id}-{type}	Separate event types for ordering
Raw TM Frames	RAW-{spacecraft_id}-{mc_count % 8}	Spread across 8 partitions per satellite

2.3 Kinesis Record Schema

Field	Type	Description
record_type	string	HK_TELEMETRY CONTACT_START CONTACT_END ANOMALY_ALERT STATE_CHANGE
schema_version	string	e.g. '1.0' — used for Lambda deserialiser versioning
ingested_at	ISO8601	UTC timestamp when record was published to Kinesis
satellite_id	string	SPACENET-1A format — matches DynamoDB satellite registry
timestamp_utc	int64	Unix epoch ms — spacecraft on-board time from CCSDS CDS
source	string	CCSDS_VC0 CCSDS_VC1 CCSDS_VC2 SYNTHETIC
payload	object	Telemetry parameters — schema varies by record_type

3. Amazon Timestream

3.1 Database & Table Design

satellite_hk	1 day	13 months	satellite_id, parameter_group_id, spacecraft_id	All spacecraft telemetry values (one measure per parameter)
orbital_state	1 day	13 months	satellite_id	altitude_km, velocity_kms, latitude_deg, longitude_deg
ground_station	1 day	3 months	station_id, contact_id	elevation_deg, link_quality_pct, rx_power_dbm
anomaly_events	7 days	13 months	satellite_id, parameter, severity	value, threshold, alert_id

3.2 Write Pattern

- Lambda consumer reads from Kinesis HK stream in batches of up to 500 records
- Each HK telemetry record is decomposed into one WriteRecord call per parameter
- Multi-measure records used where possible — up to 256 measures per write
- Timestream write endpoint accessed via VPC endpoint (no public traffic)
- Write latency target: < 100ms p99
- Scheduled queries pre-aggregate 1-minute and 1-hour rollups for Grafana performance

4. Amazon S3 — Raw Frame Archive

4.1 Bucket Design

spacenet-raw-frames-{account}	CCSDS TM raw frames	S3 Intelligent-Tiering	7 years	WORM — Compliance mode
spacenet-contact-reports	Post-contact PDF/JSON reports	S3 Standard → IA (90d)	5 years	Governance mode
spacenet-science-data	Payload science downlink files	S3 Standard → Glacier (30d)	30 years	Governance mode
spacenet-artifacts	Lambda ZIPs, EKS manifests	S3 Standard	Indefinite	None
spacenet-logs-archive	CloudTrail, VPC Flow Logs	S3 Standard → Glacier (90d)	90 years	Compliance mode

4.2 Object Key Structure

raw-frames	year=YYYY/month=MM/day=DD/sat={id}/contact={year}/{month}/{day}/sat=SN1A/contact=001/00042	year=2026/month=06/day=06/sat=SN1A/contact=001/00042
contact-reports	year=YYYY/month=MM/day=DD/sat={id}/{contact_id}	year=2026/month=06/day=06/sat=SN1A/CONTACT-2026-001
science-data	year=YYYY/month=MM/sat={id}/apid={apid}/{contact_id}	year=2026/month=06/sat=SN1A/apid=0x300/CONTACT-2026-001

4.3 Lifecycle Rules

raw-frames	S3 Standard	S3 Standard-IA	S3 Glacier IR	S3 Glacier Deep Archive

contact-reports	S3 Standard	S3 Standard-IA	S3 Standard-IA	S3 Glacier IR
science-data	S3 Standard	S3 Glacier IR	S3 Glacier IR	S3 Glacier Deep Archive
logs-archive	S3 Standard	S3 Standard-IA	S3 Glacier IR	S3 Glacier Deep Archive

5. Amazon DynamoDB

5.1 Table Design

satellites	satellite_id (S)	--	name, scid, state, launch_date, tie, On-demand	state-index (GSI on state)	
contacts	contact_id (S)	--	satellite_id, gs_id, start_time, end_time, status, frame_rate, satellite-index	status-index	
telemetry_state	satellite_id (S)	param (S)	value, unit, ts, threshold_hi, threshold_lo, On-demand	None	
commands	command_id (S)	--	satellite_id, apid, payload, status, approval, links, satellite-index	satellite-command-index	
anomalies	alert_id (S)	--	satellite_id, parameter, value, severity, resolved, On-demand	satellite-alert-index, open-index	
orbital_state	satellite_id (S)	ts (N)	lat, lon, alt, vel, az, el	On-demand — TTL 24h	

5.2 Global Tables

- contacts, telemetry_state, anomalies, commands tables enabled as DynamoDB Global Tables
- Regions: ap-southeast-2 (primary write) + us-east-1 (DR read/write)
- RPO < 10 seconds — async replication between regions
- Point-in-time recovery (PITR) enabled on all tables — 35-day window
- On-demand capacity — auto-scales to burst during contact windows

6. Amazon OpenSearch Service

6.1 Cluster Configuration

Domain name	spacenet-telemetry-analytics
Version	OpenSearch 2.13
Master nodes	3x r6g.large.search (dedicated)
Data nodes	6x r6g.2xlarge.search (2 AZs, 3 per AZ)
Storage per node	2TB gp3 EBS
Total storage	12TB hot

UltraWarm	Enabled — move indices older than 7 days to warm storage
Cold storage	Enabled — move indices older than 30 days
VPC access	Private VPC endpoint — no public access
Encryption	KMS CMK at rest + node-to-node TLS
Fine-grained access	IAM-based + internal user database for Grafana

6.2 Index Strategy

telemetry-hk-YYYY.MM.DD	Kinesis Firehose from HK stream	30 days hot, 13mo warm	3 primary, 1 replica
anomalies-YYYY.MM	EventBridge → Lambda	13 months	2 primary, 1 replica
contact-events-YYYY.MM	EventBridge contact events	13 months	2 primary, 1 replica
groundstation-YYYY.MM.DD	CloudWatch Logs subscription	30 days	2 primary, 1 replica

7. Amazon RDS Aurora PostgreSQL

Engine	Aurora PostgreSQL 16.2
Instance type	db.r7g.xlarge (writer) + db.r7g.xlarge (reader x2)
Storage	Aurora auto-scaling — starts 20GB, max 128TB
Multi-AZ	Yes — writer + 2 readers across 3 AZs
Global database	Enabled — DR replica in us-east-1 (< 1s replication lag)
Backup retention	35 days PITR
Encryption	KMS CMK
VPC access	Private subnet only — no public endpoint
Schema	mission_planning: satellites, contact_requests, tie_history, operators, audit_log

8. Data Retention & Compliance Summary

Raw TM frames	S3 + Object Lock	30 days	Glacier 365 days	7 years	Yes
HK telemetry	Timestream	1 day	Magnetic 13 months	13 months	No

Orbital state	Timestream	1 day	Magnetic 13 months	13 months	No
Contact reports	S3	90 days	IA → Glacier	5 years	Partial
Fleet state	DynamoDB	Indefinite	PITR 35 days	Indefinite	No
Commands	DynamoDB	Indefinite	PITR 35 days	7 years (audit)	Yes
Log analytics	OpenSearch	30 days	UltraWarm 13mo	13 months	No
Audit trail	S3 + CloudTrail	90 days	Glacier	7 years	Yes